SUPPLEMENTARY INFORMATION

Nested Association Mapping (NAM) in quinoa (*Chenopodium quinoa*) for yield-related traits in the Peruvian Highlands

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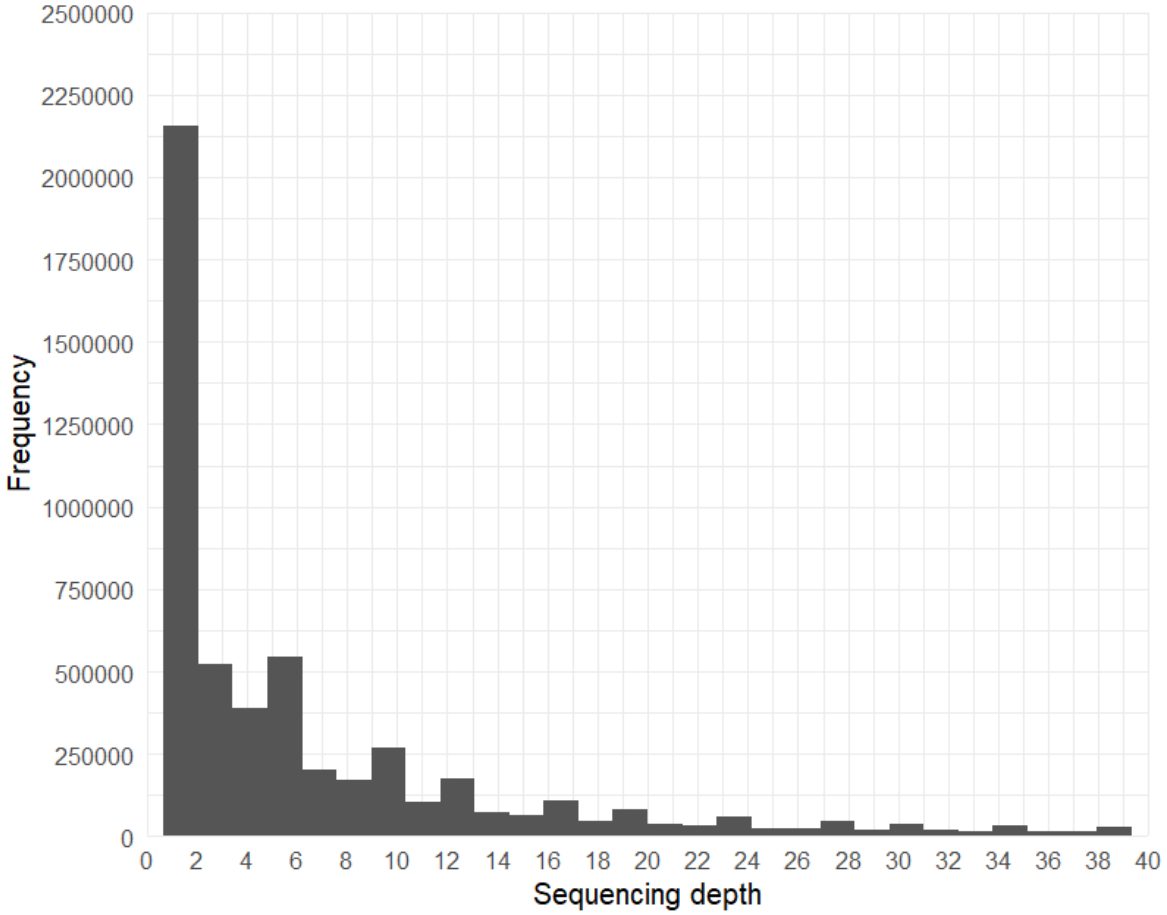
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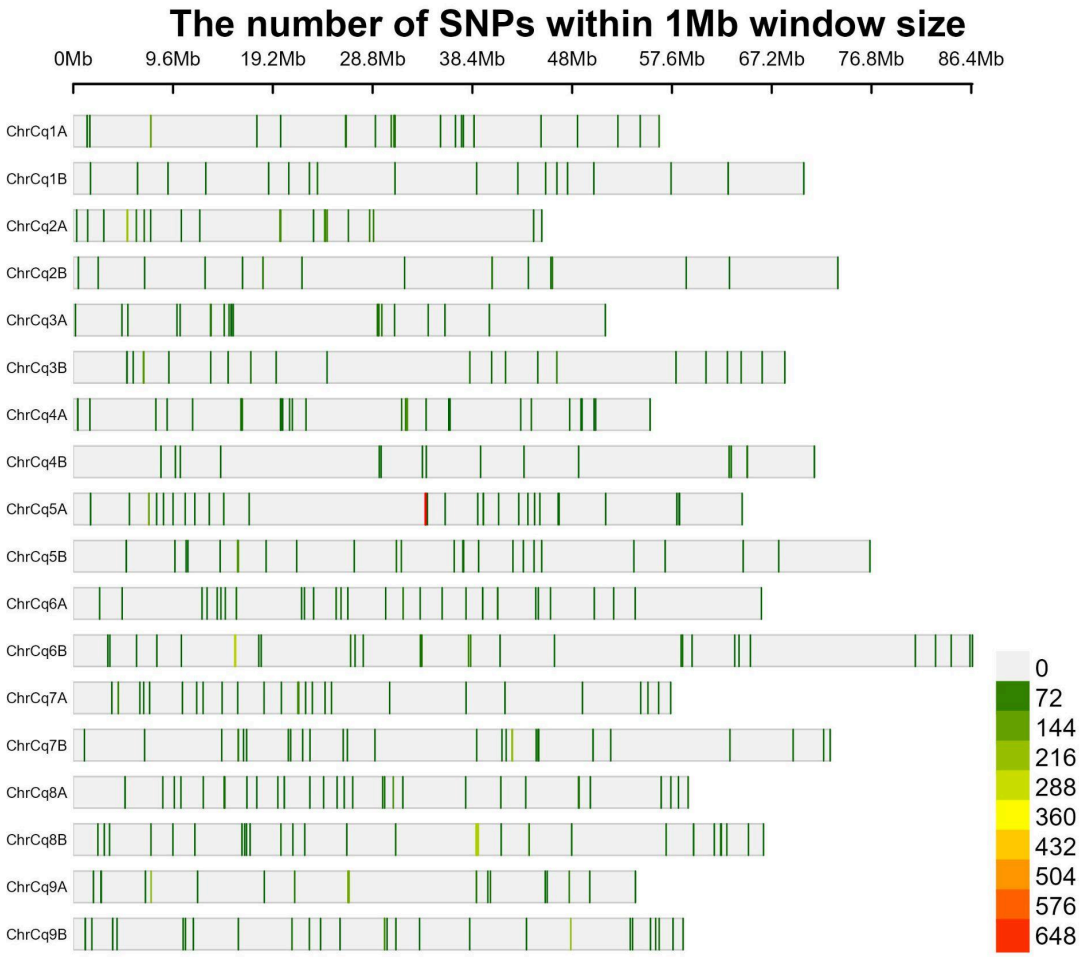
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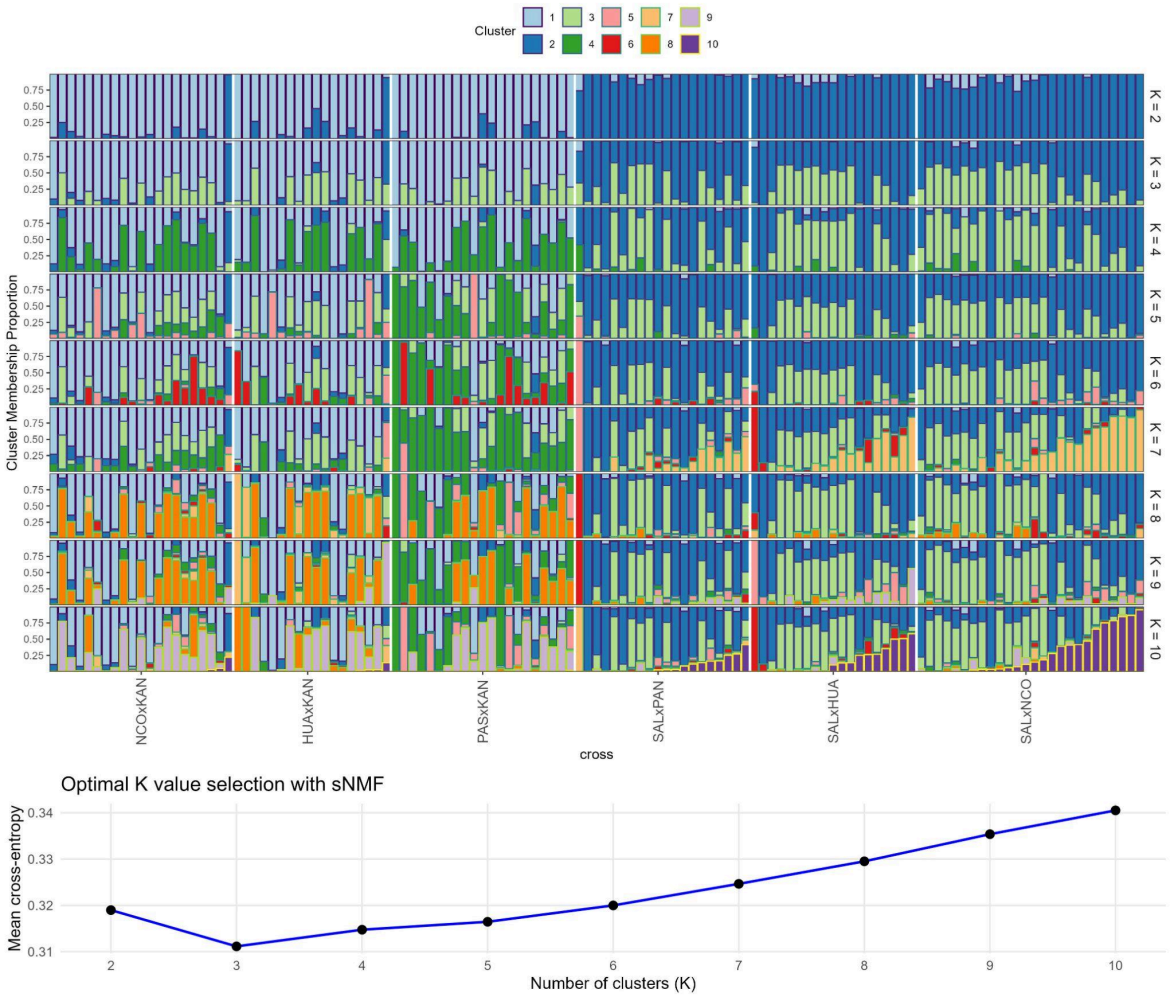
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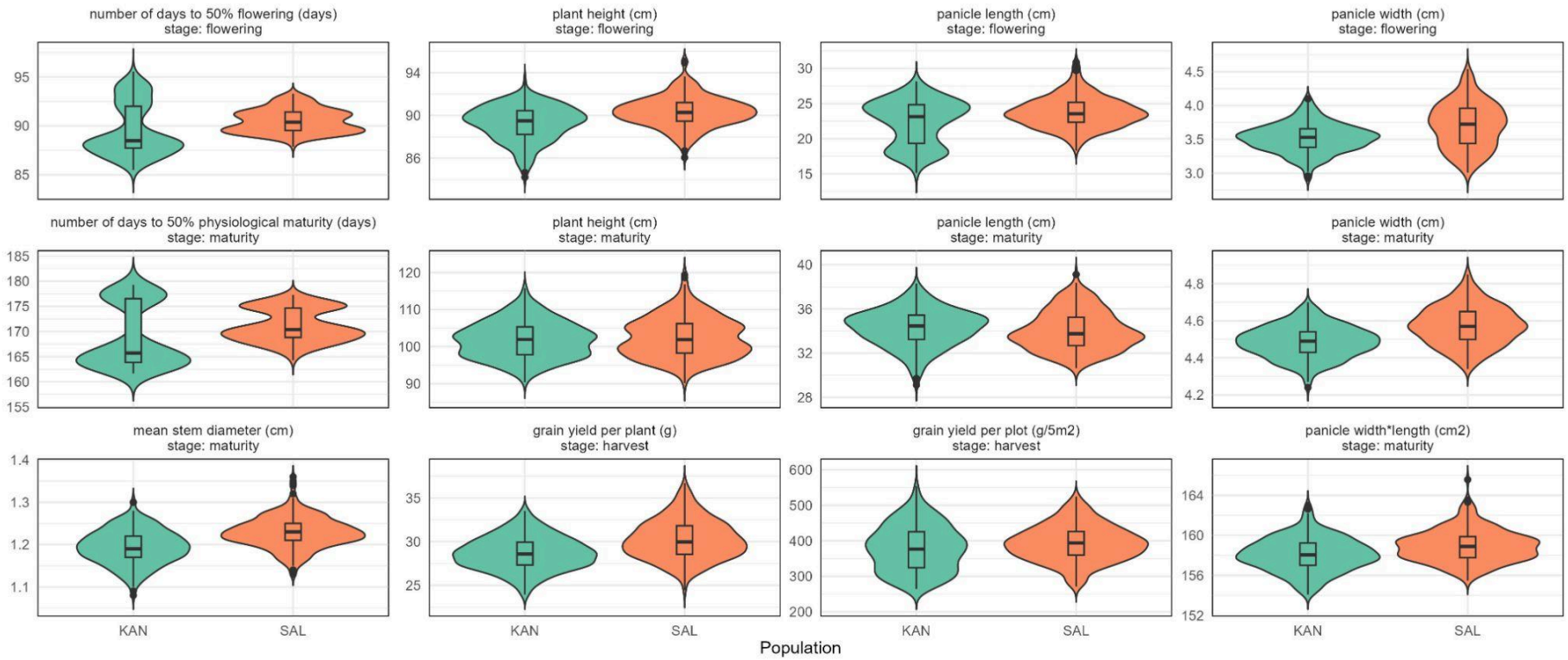
Supplementary Figure 1. Frequency distribution of sequencing depth across all SNP loci retained after variant calling.



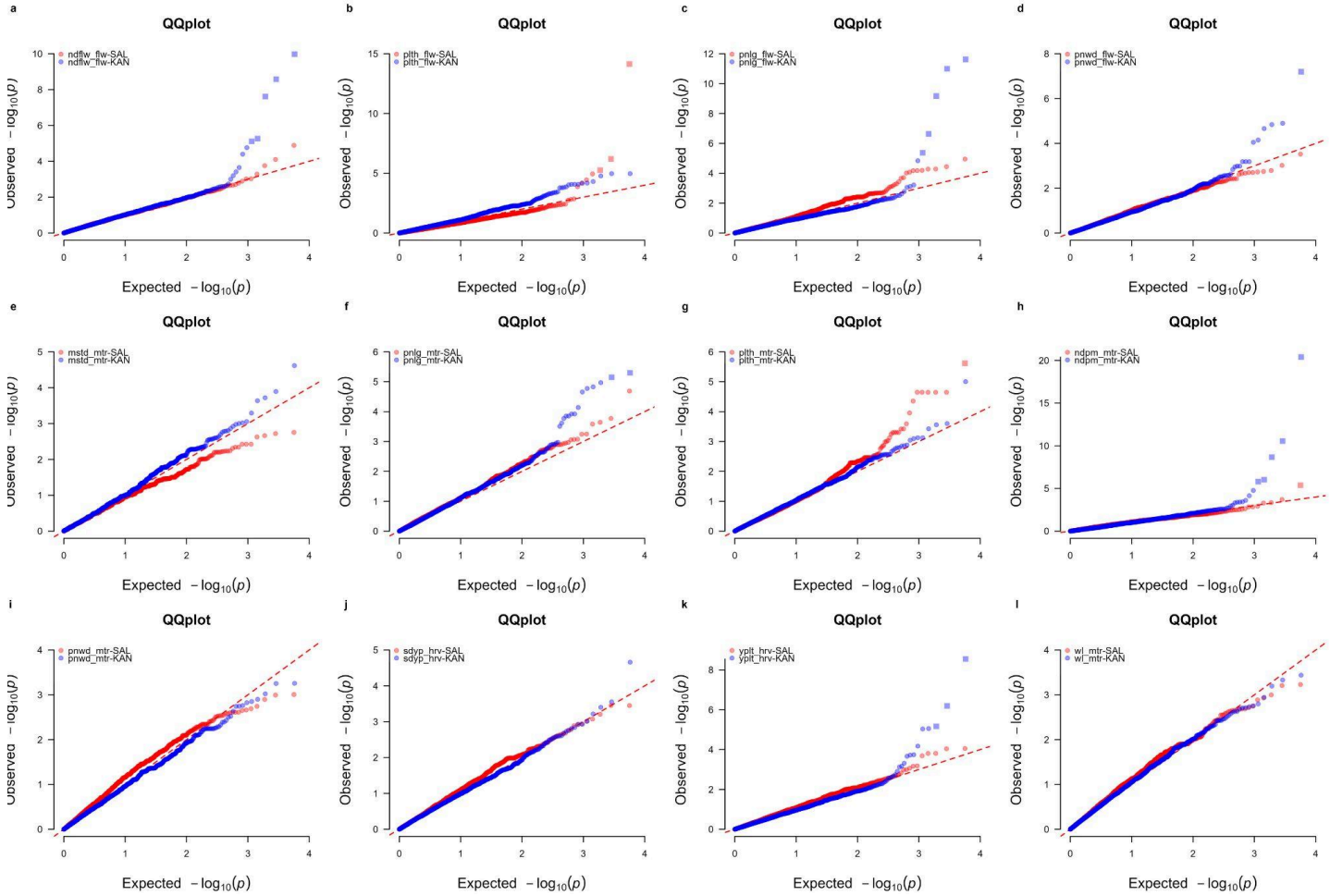
Supplementary Figure 2. Single-nucleotide polymorphisms (SNPs) distribution after filtering. Genotyping data started with 61528 SNPs, and after filtering using: bi-allelic; non-polymorphic; min_ind=0.5; facet=cross; min_loci=0.75, using the snpR R package. Final SNPs database with 6046 and 125 individuals.



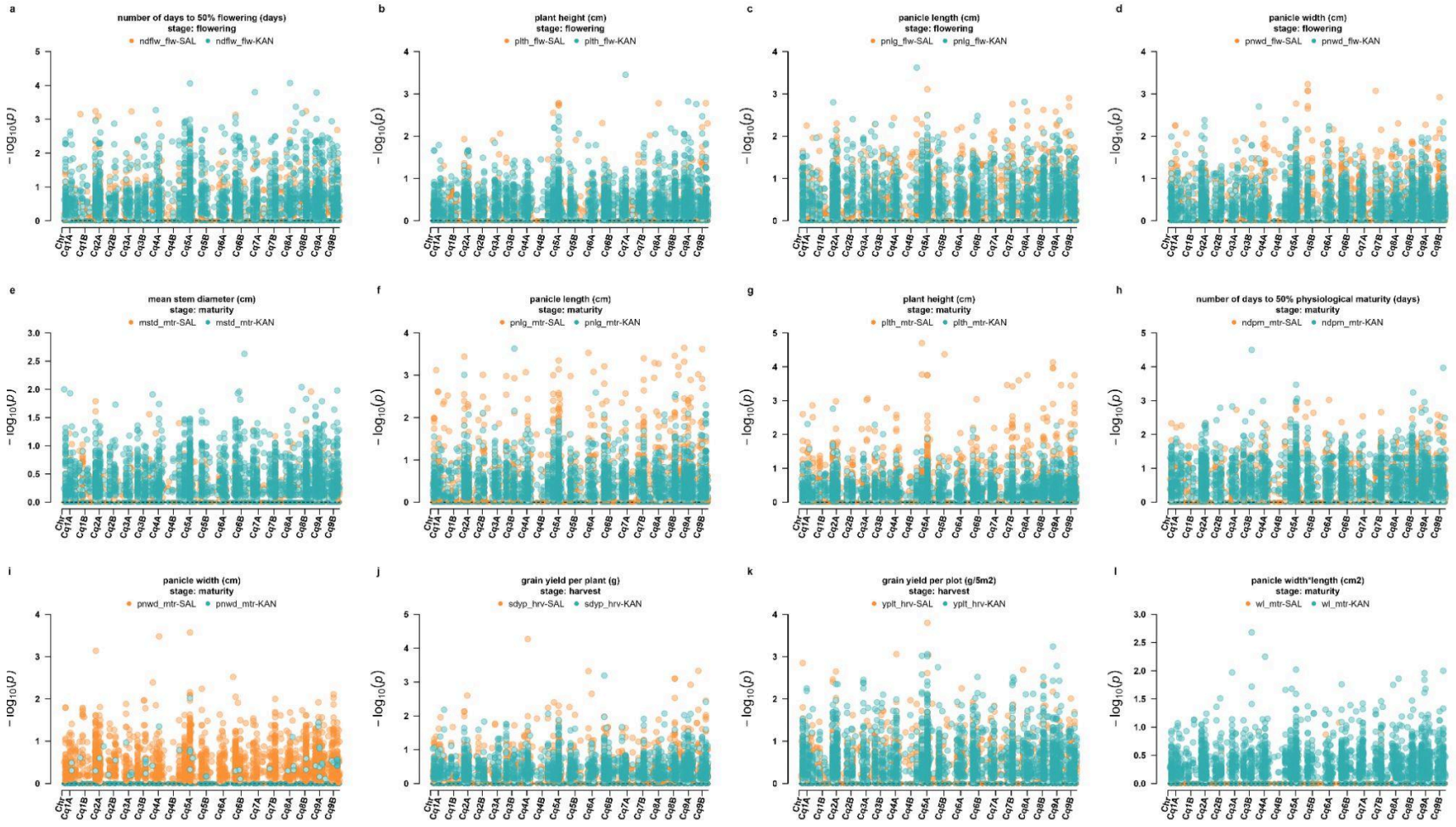
Supplementary Figure 3. Population structure inferred with sparse non-negative matrix factorization (sNMF) across six quinoa crosses. (a) Ancestry barplots for assumed ancestral clusters from K = 2–10. (b) Selection of the optimal K using the mean cross-entropy criterion shows a minimum at K = 3. Each vertical bar represents an individual, and colors indicate cluster-membership proportions; individuals are grouped by cross: Negra Collana × Kancolla (NCO×KAN), Huariponcho × Kancolla (HUA×KAN), Pasankalla × Kancolla (PAS×KAN), Salcedo INIA × Pandela Rosada (SAL×PAN), Salcedo INIA × Huariponcho (SAL×HUA), and Salcedo INIA × Negra Collana (SAL×NCO).



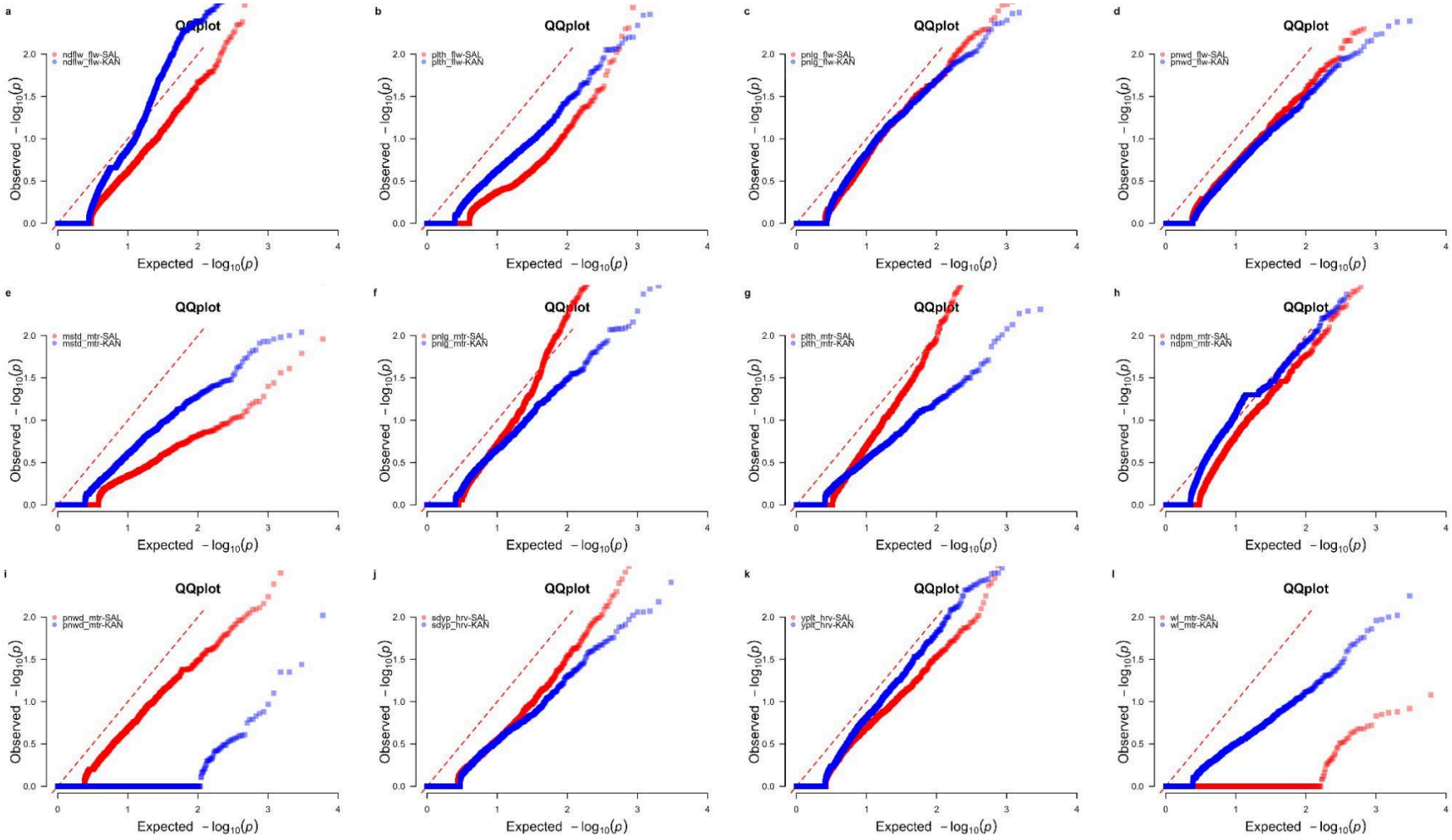
Supplementary Figure 4. Traits distribution for yield related traits in quinoa (*Chenopodium quinoa*) in two NAM populations. Kancolla (green) and Salcedo (red) NAM population under multilocation trials in the Peruvian Highlands.



Supplementary Figure 5. QQplot for Genome-Wide Association Studies in Salcedo (red) and Kancolla (blue) NAM population under multilocation trials using the *rMVP* R package (n = 125). (a) Number of days to 50% (days, ndflw), (b) Plant height (cm, plth), (c) Panicle length (cm, pnlg), (d) Panicle width (cm, pnwd), (e) Mean stem diameter (cm, mstd), (f) Number of days to 50% physiological maturity (days, ndpm), (g) Grain yield by plant (g, gypp), (h) yield per plot (g/5m2, yplt), and (i) Panicle width*length (cm2, wl).



Supplementary Figure 6. Genome-wide association study for yield related traits in quinoa (*Chenopodium quinoa*) in two NAM populations using *NAM* R package. (a) Number of days to 50% (days, ndflw), (b) Plant height (cm, plth), (c) Panicle length (cm, pnlg), (d) Panicle width (cm, pnwd), (e) Mean stem diameter (cm, mstd), (f) Number of days to 50% physiological maturity (days, ndpm), (g) Grain yield by plant (g, gypp), (h) yield per plot (g/5m2, yplt), and (i) Panicle width*length (cm2, wl). Orange and green points denote SAL- and KAN-derived NAM populations, respectively (n = 125). The horizontal line indicates the genome-wide significance at Bonferroni threshold (p-value < 0.000008).



Supplementary Figure 7. QQplot for Genome-Wide Association Studies in Salcedo (red) and Kancolla (blue) NAM population under multilocation trials using the *NAM* R package (n = 125). (a) Number of days to 50% (days, ndflw), (b) Plant height (cm, plth), (c) Panicle length (cm, pnlg), (d) Panicle width (cm, pnwd), (e) Mean stem diameter (cm, mstd), (f) Number of days to 50% physiological maturity (days, ndpm), (g) Grain yield by plant (g, gypp), (h) yield per plot (g/5m2, yplt), and (i) Panicle width*length (cm2, wl).